

REPORT 3.2 - The green flow: cup, ball, putts, score, then the shots

Opus, 2026-09-16, spawned by Fable for one job. Tree at 214624a, clean, at the spawn. Built from docs/SPEC_green-flow.md. His words that bound it:

"Is it possible for me to mark the cup, putt 1, and my score. Then the app computes the shot locations from the track and auto fill the lies and distances. Then I click on any entries to manually correct.?" (2026-09-13)

"Workflow on the green mark the cup or my ball first whatever is easiest. Hole out - record the putt length for short putts, double check GPS for long putts, enter hole score (once this is entered the app needs to compute the shots and ask me questions about the lie. Shot 2 rough or fairway, shot 3 green or fairway, etc..." (2026-09-15)

"With the Green flow established at 20ft not 15" (2026-09-16, the ruling in docs/DECISIONS_LOG.md at a778194)

1. What he does differently on the green, in five lines

1. **GREEN** in the footer (it said ENTER PUTTS) opens one sheet that is the whole green.
2. At the top of it: **MARK CUP** and **MARK BALL**, either order, each a 3 s burst that comes straight back to the sheet. No lie is asked for the ball.
3. Putts as before. **Putt 1 at 20 ft and over: the GPS distance stands** and SAVE is ready with nothing typed. **Under 20 ft the sheet will not save** until he taps the number - it shows what GPS thinks, greyed, and says so.
4. **SAVE goes straight to the score** on a hole he did not mark: strokes and penalties only, the putts already in it and not asked twice. Then FIND MY SHOTS, the lie of each shot, SAVE HOLE.
5. The ball mark and the cup **survive the hole being written**, so the first putt stays a measured putt.

A hole he *did* mark is unchanged: SAVE ends it, because his marks are the shots.

2. What changed

file	change
js/ui/screen-play.js	TYPED_PUTT_MAX_FT = 20 and firstPuttEntryMode({gpsFt}) exported; the putt-1 field and SAVE gated on them; MARK CUP + MARK BALL head row on the green sheet (the cup control left the putt-1 field); saveBall and the kind === 'putt' branch in commit; reopenPuttsAfterCup renamed reopenPuttsAfterMark and used by both; poorMarkWarning('ball') re-marks in place rather than undoing; SAVE hands an unmarked hole to openScoreCard; openHoleCard(h1, { fromSheet }); applyHoleEntry keeps green shots that carry a mark; the footer's ENTER PUTTS became GREEN
test/run.js	new group "the green flow (cup, ball, putts, score)" - 9 tests, driven through the real screen, the real sheets and a real 1 Hz pocket track in IndexedDB
test/index.html	runs the new group

Nothing else moved. Untouched: js/gps/, js/util/geo.js, js/data/schema.js (schemaVersion and formatVersion unmoved - no new keys), js/analysis/, js/round/track-analysis.js, js/round/round.js, every CSS file, the lie card and its fold, the v27 hint strings, the mark burst, the pocket lock, js/native/*, android/*.

BUILD.id is **NOT bumped**, and **neither is the** gt-shell-`<id>` **cache** in sw.js. The bump is owed at the push, on his word - the suite enforces that the two agree, so they move together in that commit. REVISION is his call as always.

3. Departures from the spec

spec line	what was built	why
4(a) TYPED_PUTT_MAX_FT "near CUP_AT_TEE_M (1094)"	at module scope, above playScreen, with his words beside it	CUP_AT_TEE_M is inside the screen's closure and cannot be exported; spec test 1 holds firstPuttEntryMode directly, so both it and the constant are module-level exports. Nothing else about the placement changed.

spec line	what was built	why
4(b) "the first green shot with no mark gets mark from reduced, else addShot"	the first green shot, marked or not, takes the mark; only a hole with no green shot at all gets a new one	spec test 4 requires "a second MARK BALL replaces the mark rather than adding a shot", which the literal reading would not do - by then the first green shot has a mark, so it would add a second. The reading both lines support is "the first green shot".
4(b) the mark built in "the same shape addShot stores"	literally: addShot builds it, and on a re-mark the mark moves onto the existing putt and the spare shot is removed	no record is constructed in the UI layer; the shape is addShot's by construction.
G3 the score card's "Penalty strokes (default 0)"	penaltyStrokes(h1), which is 0 in this state	a penalty logged when it occurred (his standing rule) is attached to a shot of this hole, and a hard 0 would throw it away. Same number in the base case.

Everything else is as written. No test is RED on purpose.

4. The suite

<http://localhost:8123/test/> (tools/devserver.py 8123, already serving this tree - confirmed with a curl of a file this job changed), browser pane fronted, viewport emulated **360 x 728** - his phone's page, REPORT 2.6.

run	tree	result
0	214624a, before any change	531 / 531
1	step 1, the 20 ft rule	535 / 535
2	step 2, MARK BALL	537 / 537
3	step 3, the score card and the marks kept	540 / 540

540 = 531 + 9. The v25-v27 fold group (spec test 7) is green on every run with no new assertion: the footer did not grow.

Every new test proven to fail against the defect first

Each mutation is one edit to js/ui/screen-play.js, run, then reverted.

mutation	what it removes	result
M1	the threshold: firstPuttEntryMode always answers gps	533 / 535 - "a tenth under it is his to type: expected typed, got gps"; "SAVE would have saved a 12 ft putt GPS cannot measure"
M2	his number: the threshold back at 15, which was Opus's recommendation and never his	533 / 535 - "20ft not 15: expected 20, got 15"; "the sheet never asked for it: 12ft"
M3	the "20 ft and over: GPS stands" line	534 / 535 - "the sheet never says GPS stands at 20 ft and over"
M4	the MARK BALL button, which is the state before step 2	535 / 537 - "the sheet has no way to mark the ball"; "no RE-MARK BALL offered once a ball is marked"
M5	the kind === 'putt' branch, so the ball takes the generic shot path	536 / 537 - "a second putt appeared out of a correction: expected 1, got 2"
M6	G4: the hole write clearing every shot	539 / 540 - "the hole write dropped the ball mark - the first putt is no longer measured"
M7	SAVE's hand-off to the score card	538 / 540 - "the sheet after SAVE was null"; "the shots stage was null"
M8	fromSheet, so the card asks the putts again	538 / 540 - "the putts were asked twice" and "the typed first putt: expected 10, got null"

M8 is the field-test-3 failure written down: a first-putt distance asked twice is a first-putt distance lost, and without it the hole contributes nothing to strokes gained putting at all.

5. What was clicked in the real app, and what it read back

?sim=1, one tab, viewport 360 x 728, the dev sim round on Veenker's back nine. n = 1 pass through the whole flow.

The footer, hole 10 with 8 shots marked, against REPORT 2.6 Section 6:

	REPORT 2.6 (v26/v27)	now
footer height	385.6 px	385.6 px
body bottom	342.4 px	342.4 px
hint	"On the green: MARK CUP, then putts."	unchanged, 1 line

The renamed button reads GREEN, 250 x 53.2 px, **0 px label spill** - G7's verify, confirmed at 360 px. Every other footer control is where it was: MARK SHOT 9 at 80 px, ENTER YARDAGES, MARK CUP, PENALTY, UNDO.

The sheet's head row on that hole - ball already marked, no cup: MARK CUP at 129.7 px and RE-MARK BALL at 126.1 px side by side, 0 px spill each, and SAVE & NEXT HOLE **disabled**: no cup, so no measurement, so the first putt is his to type.

The flow, hole 12, par 4, 306 yd, nothing marked:

step	what came back
GREEN	sheet "Hole 12 - putts", head row MARK CUP / MARK BALL, SAVE disabled
MARK BALL, burst ends	banner "Ball marked here.", no lie card , the sheet reopened, head row now MARK CUP / RE-MARK BALL
MARK CUP, burst ends	banner "Cup marked here.", the sheet reopened, head row RE-MARK CUP / RE-MARK BALL
putt 1 readout	9ft, greyed at opacity 0.45, under "GPS says 9 ft - under 20 ft, type it." - SAVE still disabled
tap 8 on the grid	readout 8ft, SAVE enabled
SAVE	the score card "Hole 12 - how did it go?", with the fields Strokes and Penalty strokes and nothing else; "2 putts saved. Now the score, and the track finds the shots."; "2 full shots for the track to find."; "Score minus putts minus penalties is what the track looks for."; FIND MY SHOTS
FIND MY SHOTS	"Hole 12 - confirm your shots. The track found 2 stops on this hole. These are the 1 most shot-like, oldest first." and the shortfall stated honestly: "1 more stroke than the track found stops... tap PLAYED TWICE on it."
PLAYED TWICE, lie Fairway, SAVE HOLE	toast "Hole 12 saved from the track."

Read back out of the store afterwards:

seq	lie	source	mark method	distance
1	tee	track	track	-
2	fairway	track	track	-
3	green	gps	burst	8 ft
4	green	manual	-	-

The cup is still there, method "burst", accuracy 1.46 m, stamped 60 s before the green entry - the cup he marked, not one recovered from the track. That is G4 in the real store: the hole was rebuilt out of the track and the two measurements he took by hand survived it.

The sim only had two stops on that hole because the round had been sitting on hole 10, so the shortfall message is the correct behaviour rather than a defect. The auto-lock fired twice during the scripted pauses - over 10 s between taps - and the sheet was still exactly where it was underneath it, which is the designed behaviour; the taps after that were driven on the elements.

6. Follow-ups, not done here

- docs/launch-checklist.md line 52 still reads "Next tee: ENTER PUTTS -> count -> paced distances". The button is GREEN now and the sheet does more than the putts. One line, not touched by this job.
- learnGreen is deliberately NOT fed by MARK BALL (spec 4(b)), so the course model still learns greens only from the lie card's marks. His call, later.
- Lies from the course map, and the "in question" edge band, wait for the course-data spec at xhigh on his word. Every lie is still his tap, and NOT SURE still flags the app's guess.

7. His call

1. **The push.** Not taken. Three commits, not pushed. BUILD.id and the gt-shell-<id> cache are owed in the push commit.
2. **The footer label GREEN (G7) and the score card's default strokes, par or the putts plus one, whichever is more (G3)** were Fable's defaults and are his to change. On a par 4 with two putts the card opens at 4.
3. **One pass on the phone** when a build ships: on the green, GREEN, MARK BALL, MARK CUP, the putts, SAVE, the score, FIND MY SHOTS. The numbers to send back if anything is wrong are what the putt 1 readout said and whether SAVE was tappable.